

- 5 Fig. 5.1 shows the view along the axis of a solenoid. The solenoid is 0.30 m in length and has 50 turns. It has a current of 2.0 A flowing through it.

An electron located at the centre of the solenoid is moving to the right.

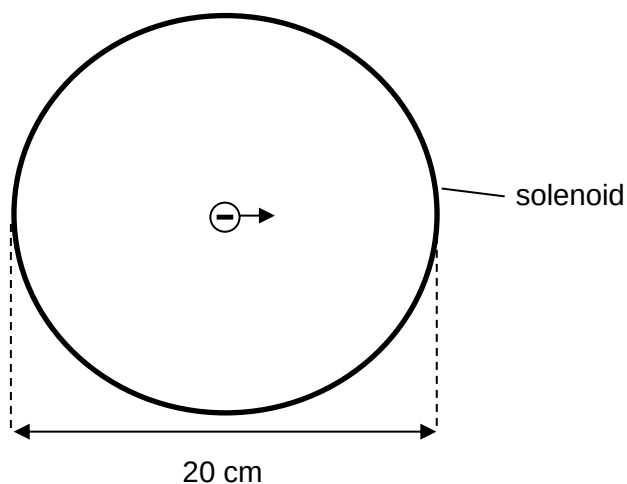


Fig. 5.1

- (a) (i) State what is meant by *magnetic flux density*.

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 [1]

- (ii) Indicate on Fig. 5.1, the direction of current flowing through the solenoid for the electron to experience a downward force.

[1]

- (iii) Calculate the magnetic flux density B at centre of solenoid.

magnetic flux density = T [1]

- (iv) Hence, determine the magnetic force acting on the electron if it is moving with a speed of 10 m s^{-1} .

force = N [2]

- (v) Explain why the force experienced by the electron due to the magnetic force does not change the speed of the electron.

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..... [2]

- (b) A uniform electric field is now applied in the region of magnetic field within the solenoid so that the electron can move to the right undeflected.

- (i) Calculate the magnitude of the electric field strength E .

electric field strength = V m^{-1} [2]

- (ii) Indicate on Fig. 5.1 with an arrow, the direction of the electric field. [1]